

# Global Electronics Retailer — Business Analysis Report

## 1. Project Overview

The Global Electronics Retailer project is an end-to-end Business Analyst and Data Analytics project focused on understanding sales performance, customer behavior, product performance, geographic performance, and the major drivers behind a significant revenue decline.

The analysis was performed using:

- **PostgreSQL** for data storage and validation
- **SQL** for business analysis and investigation
- **Power BI** for interactive dashboard development
- **Business Analysis techniques** for identifying findings, root causes, and management recommendations

## 2. Business Problem

The business experienced strong revenue growth from 2016 through 2019 but faced a significant decline in sales during 2020.

The primary objective was to answer:

1. How is the business performing overall?
2. Which products and categories generate the most revenue?
3. Which countries and channels are most important?
4. What happened to sales during the 2020 decline?
5. Was the decline caused by fewer customers, lower purchase frequency, lower order value, or fewer orders?
6. Which markets and product categories were most affected?
7. What actions should management prioritize?

## 3. Dataset

The project contains six source tables:

- customers
- products
- sales
- stores
- exchange\_rates
- data dictionary

### ➤ Main dataset size

| Table           | Rows   |
|-----------------|--------|
| Customers       | 15,266 |
| Products        | 25,180 |
| Sales           | 62,884 |
| Stores          | 67     |
| Exchange Rates  | 11,215 |
| Data Dictionary | 37     |

## 4. Data Model

The Power BI model uses a relational structure connecting the main dimension tables to the sales fact table.

### Relationships

- `customers[CustomerKey] → sales[customerkey]`
- `products[productkey] → sales[productkey]`
- `stores[storekey] → sales[storekey]`

The relationships are configured as **one-to-many** with single-direction filtering.

The `exchange_rates` table was kept disconnected because currency conversion requires both **date** and **currency code** rather than a simple single-column relationship.

## 5. Revenue Calculation

The sales table does not contain a direct revenue column.

Revenue was calculated using:

**Revenue = Quantity × Product Unit Price**

SQL and Power BI calculations were cross-checked to ensure consistent results.

## 6. Key Business KPIs

| KPI              | Result          |
|------------------|-----------------|
| Total Sales      | <b>\$55.76M</b> |
| Total Orders     | <b>26,326</b>   |
| Total Quantity   | <b>197,757</b>  |
| Total Customers  | <b>15,266</b>   |
| Active Customers | <b>11,887</b>   |

|                                    |                   |
|------------------------------------|-------------------|
| Average Order Value                | <b>\$2,117.80</b> |
| Average Quantity per Order         | <b>7.51</b>       |
| Sales per Active Customer          | <b>\$4,690.46</b> |
| Repeat Customers                   | <b>7,272</b>      |
| Customer Purchase Rate             | <b>77.88%</b>     |
| Repeat Customer Rate               | <b>61.17%</b>     |
| Average Orders per Active Customer | <b>2.21</b>       |

## 7. Business Analysis Findings

### Finding 1 — Top Products

The top five products generated approximately **\$2.32M**, representing only **4.16%** of total sales.

All five leading products were desktop PC variants.

#### Insight

- Desktop products are strong performers, but revenue is not excessively dependent on a small number of individual products.

#### Recommendation

Maintain strong inventory availability for high-performing desktop products while continuing to diversify the product portfolio.

### Finding 2 — Category Revenue Concentration

Computers generated:

**\$19.30M — 34.62% of total sales**

Home Appliances was the second-largest category at:

**\$10.80M — 19.36%**

#### Insight

- Computers generate nearly one-third of company revenue and are approximately 1.78 times larger than the second-largest category.

## Recommendation

Protect inventory, availability, and promotional support for Computers while continuing to develop other high-performing categories to reduce dependency on one category.

## Finding 3 — Geographic and Channel Concentration

The United States generated:

**\$23.76M — 42.62% of total sales**

Online generated:

**\$11.40M — 20.45%**

Together, the U.S. and Online segment represented:

**63.07% of total revenue**

### Insight

- Revenue is highly concentrated in the U.S. and Online channel.

## Recommendation

Protect these core revenue drivers while developing lower-performing markets to improve geographic diversification.

## Finding 4 — Revenue Trend

Revenue increased strongly from 2016 through 2019.

Approximate annual revenue:

- 2016: **\$6.95M**
- 2017: **\$7.42M**
- 2018: **\$12.79M**
- 2019: **\$18.26M**
- 2020: **\$9.29M**

Revenue declined approximately **49% from 2019 to 2020**.

The major turning point occurred during **March–April 2020**.

## Recommendation

Investigate March–April 2020 as the critical disruption period and establish monthly performance monitoring to identify future declines earlier.

## Finding 5 — Order Volume Collapse

The analysis showed that the 2020 revenue decline was primarily associated with a collapse in transaction volume.

Between February and April 2020:

- Orders declined from **1,124 to 80**
- Order volume declined **92.9%**
- Sales declined from **\$2.23M to \$217.6K**
- Sales declined **90.2%**

## Insight

- The magnitude of the order decline is much larger than the decline in customer activity.

## Recommendation

Prioritize investigation and recovery of transaction volume rather than focusing primarily on increasing order value.

## Finding 6 — Market-Level Impact

March–April 2020 compared with March–April 2019:

| Market         | Change |
|----------------|--------|
| Germany        | –72.1% |
| Australia      | –76.5% |
| Netherlands    | –78.1% |
| France         | –41.1% |
| United States  | –3.3%  |
| Online         | +19.6% |
| United Kingdom | +31.2% |
| Canada         | +44.7% |
| Italy          | +32.2% |

## Insight

- The revenue shock was highly uneven across markets.

- Germany, Australia, and the Netherlands experienced severe declines, while Online, the UK, Canada, and Italy showed resilience or growth.

## Recommendation

Prioritize root-cause investigation in Germany, Australia, and the Netherlands while studying resilient markets to identify successful practices that can be replicated.

## Finding 7 — Product Category Impact

March–April 2020 compared with March–April 2019:

| Category                      | Change |
|-------------------------------|--------|
| Home Appliances               | –44.8% |
| Audio                         | –34.9% |
| Computers                     | –9.6%  |
| Games and Toys                | –7.0%  |
| Music, Movies and Audio Books | –5.4%  |
| TV and Video                  | –1.4%  |
| Cell phones                   | +6.8%  |
| Cameras and camcorders        | +33.7% |

## Insight

The decline was concentrated in specific categories rather than affecting every category equally.

Home Appliances and Audio experienced significant declines, while Cameras and Camcorders and Cell Phones remained resilient.

## Recommendation

Investigate inventory, supply, pricing, and demand conditions in Home Appliances and Audio while studying resilient categories for transferable strategies.

## Finding 8 — Active Customer Decline

Active customers during March–April:

- 2019: 456
- 2020: 407

Change:

–10.8%

## Insight

- The active customer base declined, but the decline was far smaller than the decline in orders. Therefore, customer loss alone cannot explain the 2020 revenue collapse.

## Recommendation

Focus the investigation beyond customer acquisition and examine transaction availability, market conditions, and operational factors.

## Finding 9 — Purchase Frequency

Orders per active customer:

- March–April 2019: **1.011**
- March–April 2020: **1.012**

The metric remained essentially unchanged.

## Insight

- Customers who were active did not significantly change their purchase frequency.

## Recommendation

Do not treat reduced purchase frequency as the primary explanation for the sales decline. Focus on the factors that reduced the overall number of transactions.

## Finding 10 — Average Order Value

Average Order Value:

- March–April 2019: **\$2,160.13**
- March–April 2020: **\$2,166.58**

Change:

**+0.3%**

## Insight

- Average spending per order remained stable.

Therefore, the sales decline was not caused by customers spending substantially less per transaction.

## **Recommendation**

Management should prioritize restoring transaction volume rather than focusing primarily on increasing average order value.

# **8. Root Cause Analysis**

The combined analysis indicates that the primary driver of the 2020 revenue decline was:

## **Order Volume Collapse**

The decline was further concentrated by:

- Specific geographic markets
- Specific product categories
- Severe disruption during March–April 2020

At the same time, the analysis found that:

- Average Order Value remained stable
- Purchase frequency remained stable
- Active customers declined by only 10.8%

## **Overall conclusion**

The evidence suggests that the business problem was primarily related to **transaction volume and market/category disruption**, rather than a major reduction in customer spending per order.

# **9. Management Recommendations**

## **➤ Priority 1 — Restore Order Volume**

Investigate the operational, market, channel, and demand factors that caused the severe reduction in transactions during March–April 2020.

## **➤ Priority 2 — Recover Weak Markets**

Prioritize Germany, Australia, and the Netherlands because they experienced the largest market-level declines.

## **➤ Priority 3 — Investigate Weak Categories**

Review Home Appliances and Audio for potential inventory, supply, pricing, or demand issues.

➤ **Priority 4 — Protect Core Revenue Drivers**

Maintain strong availability and commercial support for:

- Computers
- United States
- Online channel

➤ **Priority 5 — Learn From Resilient Markets**

Study Online, UK, Canada, and resilient product categories to identify practices that helped maintain or increase sales.

➤ **Priority 6 — Diversify Revenue**

Reduce long-term concentration risk by expanding stronger performance across additional markets and product categories.

➤ **Priority 7 — Improve Monitoring**

Establish regular monitoring of:

- Monthly sales
- Order volume
- Active customers
- Average Order Value
- Category performance
- Country performance

This would allow management to identify significant changes earlier.

# 10. Power BI Dashboard

The final Power BI dashboard contains three pages

## 1. Page 1 — Business Performance



Figure 1 Executive Sales Overview

Focuses on:

- Sales KPIs
- Order KPIs
- Customer KPIs
- Sales trends
- Product categories
- Countries
- Stores
- Brands
- States
- Product subcategories

## 2. Page 2 — Product & Customer Analysis

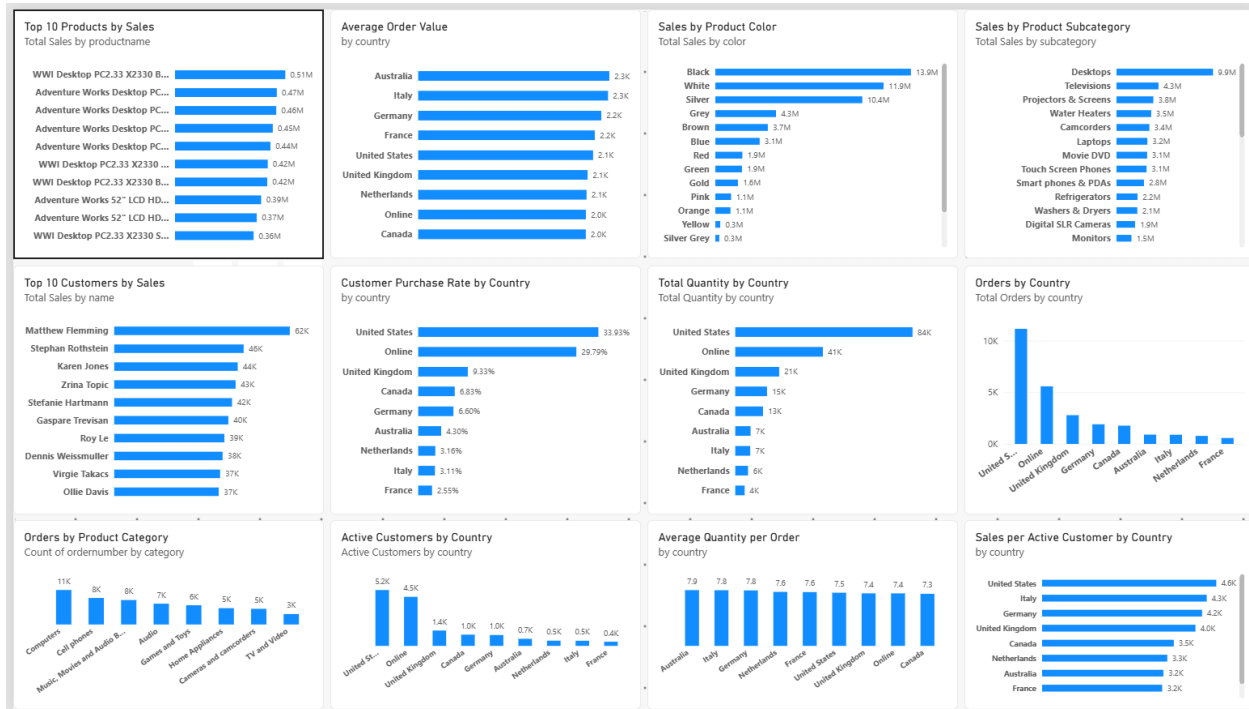


Figure 2 Product & Customer Analysis

Focuses on:

- Top products
- Top customers
- Orders by category
- Average Order Value by country
- Customer purchase rate
- Active customers
- Product colors
- Quantity by country
- Average quantity per order
- Product subcategories
- Orders by country

### 3. Page 3 — Business Insights & Actions

| Executive Summary                                                                                                                                                                                                                                     |                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Sales declined 49% in 2020, mainly due to a sharp collapse in order volume during March–April. The decline was concentrated in specific markets and categories, while customer spending per order remained stable.                                    |                                                                                                    |
| <b>ORDER VOLUME — PRIMARY ISSUE</b>                                                                                                                                                                                                                   | <b>MARKET IMPACT</b>                                                                               |
| Orders fell 92.9% from February to April 2020.<br>Action: Focus on restoring transaction volume.                                                                                                                                                      | Germany –72%   Australia –77%   Netherlands –78%<br>Action: Investigate the most affected markets. |
| <b>CATEGORY IMPACT</b>                                                                                                                                                                                                                                | <b>CUSTOMER BEHAVIOR</b>                                                                           |
| Home Appliances –45%   Audio –35%<br>Action: Review inventory, supply, pricing and demand.                                                                                                                                                            | Active Customers –11%   AOV +0.3%<br>Action: Focus on transaction volume, not order value.         |
| <b>RESILIENT MARKETS</b>                                                                                                                                                                                                                              | <b>REVENUE CONCENTRATION</b>                                                                       |
| Online +20%   UK +31%   Canada +45%<br>Action: Identify and replicate successful practices.                                                                                                                                                           | Computers 34.6%   U.S. 42.6%<br>Action: Protect core revenue drivers and diversify.                |
| <b>ROOT CAUSE</b>                                                                                                                                                                                                                                     |                                                                                                    |
| Revenue decline was primarily driven by a collapse in order volume, with the largest impact coming from specific markets and product categories.<br>• AOV and purchase frequency remained stable, so lower spending per order was not the main issue. |                                                                                                    |
| <b>MANAGEMENT PRIORITIES</b>                                                                                                                                                                                                                          |                                                                                                    |
| 1. Recover: Germany, Australia & Netherlands<br>2. Investigate: Home Appliances & Audio<br>3. Protect: Computers, U.S. & Online<br>4. Learn: UK, Canada & resilient categories<br>5. Monitor: Orders, sales & customer activity                       |                                                                                                    |
| <b>FINAL RECOMMENDATION</b>                                                                                                                                                                                                                           |                                                                                                    |
| Restore order volume first. Investigate the most affected markets and categories, protect strong revenue drivers, and replicate successful practices from resilient markets and channels.                                                             |                                                                                                    |

Figure 3 Insights & Recommendations

Summarizes:

- Revenue decline
- Order volume collapse
- Market impact
- Category impact
- Customer behavior
- Resilient markets
- Revenue concentration
- Root cause
- Management priorities
- Final recommendation

## 11. Final Business Recommendation

**Restore order volume first.**

Management should investigate the severe transaction decline, particularly in the most affected markets and product categories. At the same time, the company should protect its strongest revenue drivers, including Computers, the U.S. market, and online sales.

Resilient markets and categories should be studied to identify successful practices that can be replicated, while longer-term diversification should reduce dependence on a limited number of markets and categories.

## 12. Tools Used

- PostgreSQL 18.6
- pgAdmin 4
- SQL
- Microsoft Power BI
- DAX
- GitHub

## 13. WORK FLOW

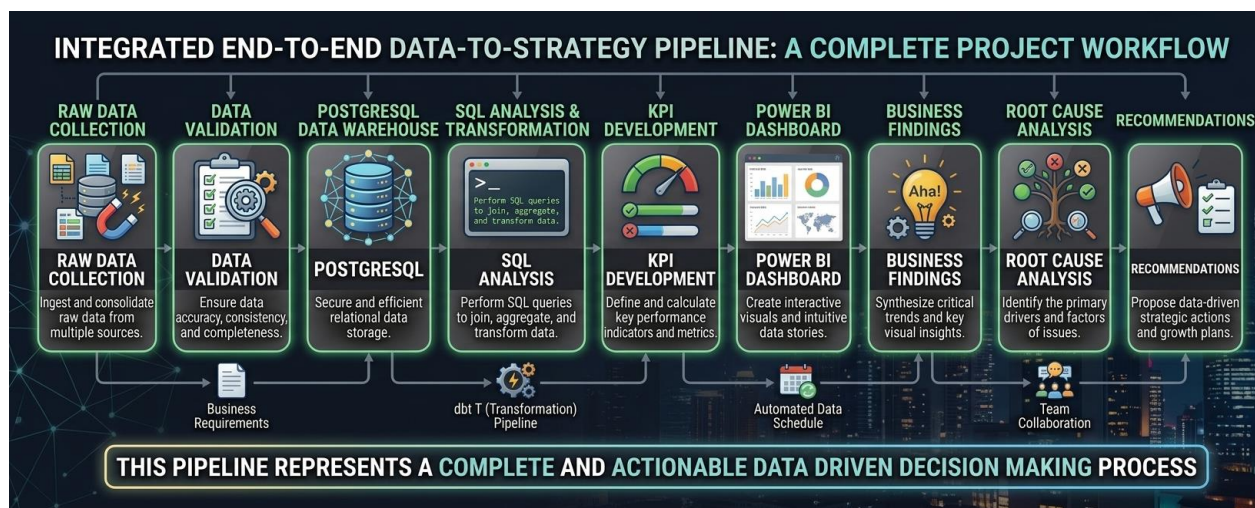


Figure 4 INTEGRATED END-TO-END DATA-TO-STRATEGY PIPELINE: A COMPLETE PROJECT WORKFLOW

## 14. Project Outcome

This project demonstrates an end-to-end Business Analyst workflow:

**Raw Data → Data Validation → PostgreSQL → SQL Analysis → KPI Development → Power BI Dashboard → Business Findings → Root Cause Analysis → Recommendations**

The project focuses not only on reporting what happened, but also on explaining **why performance changed and what management should do next.**